Agenda



- Introduction
- Approaches for Managing Customer Heterogeneity
  - Evolution of Approaches for Managing Customer Heterogeneity
  - Segmenting, Targeting, and Positioning (STP) Approach
  - Customer-Centric Approach
- **Framework for Managing Customer Heterogeneity**
  - Inputs to Managing Customer Heterogeneity Framework
  - Outputs of Managing Customer Heterogeneity Framework
  - Process for Managing Customer Heterogeneity
- Takeaways
- Examples/Case

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Marketing Principle #1: All Customers Differ
➔ Managing Customer Heterogeneity



Inputs (3Cs)

All Potential Customers
- Needs
- Demographics
- Size, growth, perceptions

Your Company
- Strengths and weaknesses
- Opportunities and threats

Your Competitors
- Strengths and weaknesses
- Opportunities and threats

Managing Customer Heterogeneity

Approaches & Processes
Segmenting, targeting, and positioning (STP)
Perceptual/positional maps
Customer-centric view

Analyses
Factor analysis
Cluster analysis
GE matrix
Discriminant analysis
Classification

Outputs (STP)

Industry Segmentation
- Customer segments
- Needs, demographics, and opportunity of each segment

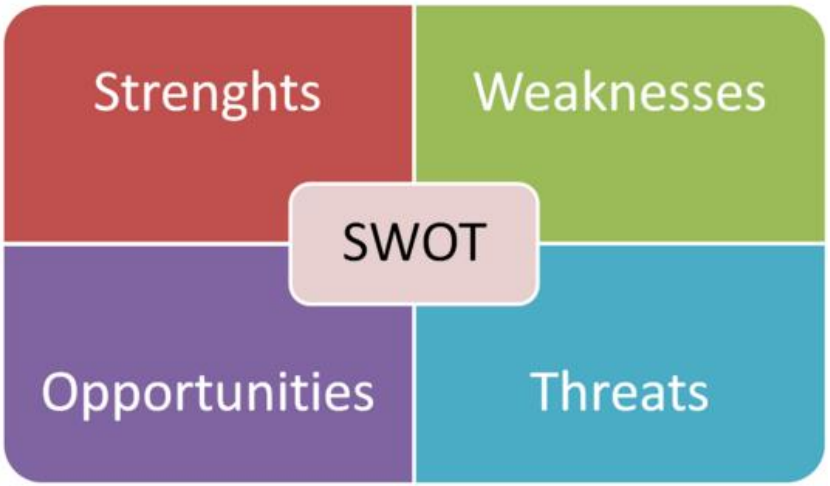
Target Segment
- Detailed needs, demographics, and value of target segment(s)
- Discriminant function
- Relative perceptions

Positioning Statement
- Who (target segments)
- What needs/benefits
- Why (relative advantage & support)

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Inputs to the Managing Customer Heterogeneity Framework

1. All potential **c**ustomers -- needs, desires, and preferences across customers in an industry, geographic region, market segment, or product category
2. Your **c**ompany -- an inventory of the company's **s**trengths, **w**eaknesses, **o**pportunities, and **t**hreats (SWOT analysis)
3. Your **c**ompetitors -- an inventory of your competitor's **s**trengths, **w**eaknesses, **o**pportunities, and **t**hreats (SWOT analysis)



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Data Analytics Technique 2.3 SWOT and 3C Analyses

Description

A SWOT appraises the *strengths*, *weaknesses*, *opportunities*, and *threats* that affect a company's success. The 3C analysis evaluates customers, competitors, and the company.

When to Use It

- To assess strategic marketing decisions by identifying critical internal and external environmental factors that will contribute to the success or failure of a strategy.
- A SWOT analysis assesses the internal and external nature of the business, across current and future situations.
- The 3C analysis focuses on these three perspectives to gain competitive advantages.

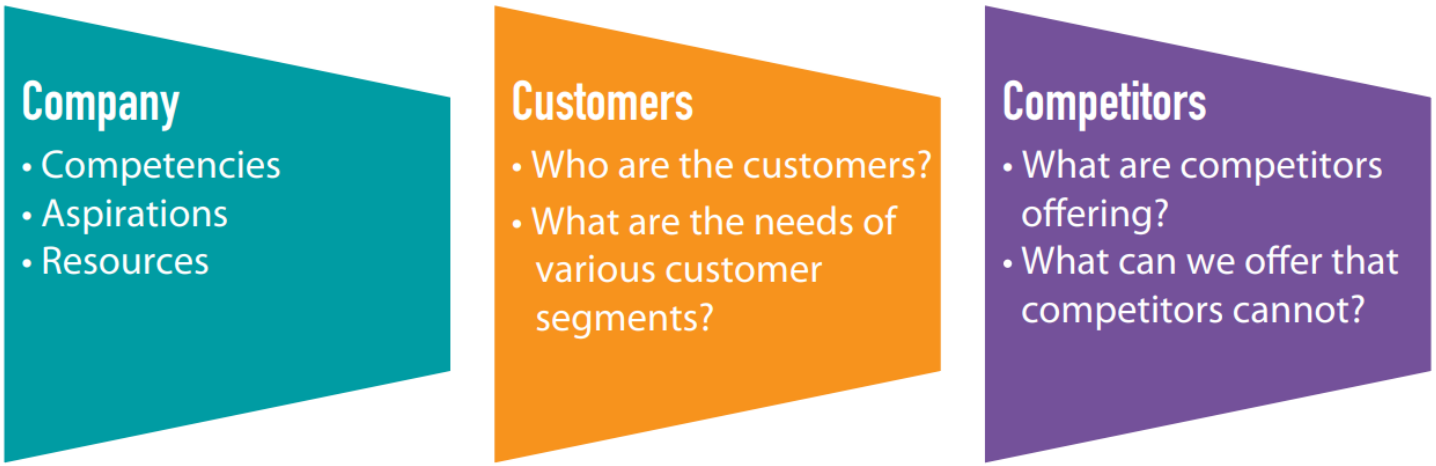
Inputs

- **External (environmental) factors**: relevant legal structure, competitors' core competencies and market share, changes in customer demographics
- **Internal (company-level) factors**: core competencies, market share, competitive advantages

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SWOT Analysis

Internal		
	Explanation	**Strategy Implications**
Strengths	Current strengths, such as a strong financial performance or a reputed brand	The firm can develop new products to leverage these strengths
Weaknesses	Current weaknesses, such as a slow customer response rate	Strategies need to be implemented to eliminate these weaknesses
External		
	Explanation	**Strategy Implications**
Opportunities	Future opportunities, such as environmental factors that may work in the company's favor	Strategies need to be devised to take advantage of the potential opportunities
Threats	Future threats, such as increasing competition	Strategies need to be devised to overcome the threats, such as lowering prices or increasing promotions



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Example

The managers of a bakery want to open a new store in a neighborhood across town. They perform SWOT and 3C analyses of the environment to assess the obstacles they may face.

SWOT Analysis

Strengths	Weaknesses
Differentiated by well-known, house-baked sourdough	New store location has limited parking
Opportunities	**Threats**
Grocery store opening next to bakery, drawing in many new potential customers	Competition is well established and has a loyal, local customer base

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Example

The managers of a bakery want to open a new store in a neighborhood across town. They perform SWOT and 3C analyses of the environment to assess the obstacles they may face.

3C Analysis

Company	Customers	Competitors
The *company* itself is known for its house-baked sourdough bread. Production costs are very high though	Potential *customers* in the new neighborhood are primarily high-income families with small children	*Competitors* in the neighborhood include a donut shop and a café with a selection of locally baked goods

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Outputs of Managing Customer Heterogeneity Framework



1. **Industry Segmentation** describes industry segments and includes, for each named segment, salient purchase preferences, demographic variables, and potential demand opportunities
   - How can the marketplace be described using homogenous groups?
   - What does each group of potential customers want?

2. **Target Segmentation** moves from the overall market landscape to the specific segment(s) of interest, such that it extends the first output by providing a very detailed description of each target segment.
   - What set of segments will the firm pursue?
   - How does the firm identify each group of target customers?

3. **Positioning Statements** encapsulate the three questions into one concise statement that firms use to direct their internal and external marketing activities: who should the firm target, what needs and benefits are being fulfilled, and why does this offering provide a relative advantage over competitive offerings

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Data Analytics Technique 2.4

Discriminant and Classification Analyses

Description

Discriminant and classification analyses are multivariate statistical techniques used to determine how segments of consumers differ in their characteristics.

When to Use It

- To classify a large set of customers into small subgroups with different characteristics.
- To predict or classify which subgroup a new customer belongs to and thus target marketing activities better.

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How It Works

Discriminant function analysis describes which predictor variables will differentiate two or more segments of customers. Assume that a prior cluster analysis has given the firm a good understanding of how many segments of customers it has. With a cluster analysis, the firm also can classify each customer to a segment, such as the quality segment versus the value segment. However, discriminant analysis usually follows cluster analysis. Managers using discriminant analysis collect variables about customers (e.g., demographics, preferred marketing channels) and describe why a customer falls in a certain segment.

Thus, the dependent variable for discriminant analysis is a categorical variable (i.e., assigned segment of a customer), and the independent variables are customer characteristics (e.g., demographics, preferred marketing channels). As an equation, a discriminant analysis is:

$$Y_1(x) = \beta_{11} \cdot x_1 + \ldots \beta_{1k} x_k$$

$$Y_h(x) = \beta_{h1} \cdot x_1 + \ldots \beta_{hk} x_k$$

In these equations, the firm has h segments, and any $Yi(x)$ is a binary variable equal to 1 if a customer belongs to the ith segment, and 0 otherwise. Because the firm uses k profiling variables, every xi denotes an independent variable. The outputs of the analysis are the weights bij that capture the influence the jth independent variable has for categorizing a customer into the ith segment. If we have h total segments and k total independent variables, we have $h \times k$ total weights.

The weights are similar to regression weights; the higher the value of a certain weight bi, the stronger the association between the corresponding predictor xi and segment membership. The real usefulness of discriminant analysis arises when a firm encounters a customer who has not been subject to cluster analysis. The firm observes xi; knowing that some xs are more likely to be associated with membership in certain segments, it can classify the customer into a segment, even without performing another cluster analysis.

Example

A company conducted a cluster analysis of 2,000 customers and found two dominant segments. Customers in one segment (quality) have very high preferences for quality and do not mind paying a high price, but customers in another cluster are very value conscious and refuse to pay high prices (value segment). To profile these segments, the company conducts a discriminant analysis using two measures: number of years in business with the customer (X_1) and market value (X_2). The estimated linear score functions for each segment are:

Quality segment: $0.52x1 - 0.58x2$
Value segment: $-0.12x1 + 1.25x2$

Thus, the company can better interpret each segment. The quality segment represents customers with small market value (according to the negative coefficient –0.58) who have been in business with the company for a long time (positive coefficient 0.52). The value segment instead represents customers with large market value (large positive coefficient 1.25) and who have been in business with the company for a short time (negative coefficient –0.12). The company thus can identify Segments 1 and 2 and classify customers simply by observing their market value and the length of their relationship.

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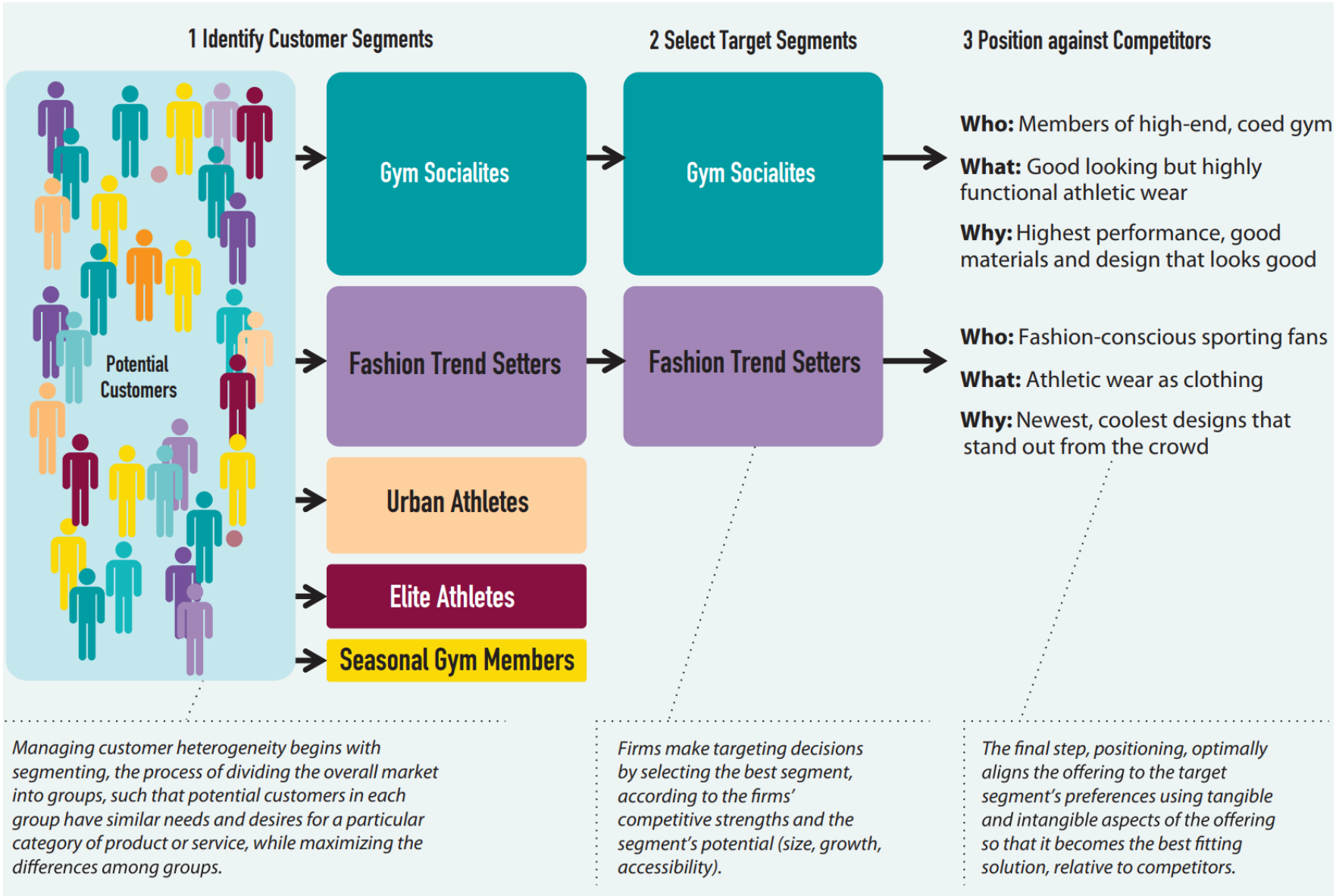
Process for Managing Customer Heterogeneity



- To convert the inputs into outputs, marketers conduct a series of process steps
  1. **Segmenting** – To initiate the segmentation, managers need to *identify the key purchase attributes*, that is the needs and desires that a potential customer evaluates when making a purchase decision for this category
  2. **Targeting** – The targeting process follows naturally from segmentation, *to identify which segments the firm wants to sell to, based on the attractiveness of each segment and the firm's competitive strength in each segment*
  3. **Positioning** – The separation between targeting and positioning is often blurry. Many of the factors used to evaluate competitive strengths to select a target segment also impact the difficulty of executing an effective positioning strategy for that segment
  4. **Building Customer Centricity** – Building a customer-centric organization is different from executing an STP process, in that it requires a top-down, enduring commitment from senior leaders to institute a customer-centric philosophy across the firm's entire organization

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Example of Managing Customer Heterogeneity



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Marketing Engineering Demo: Steps for STP



1. Check survey data (reasonable means, ranges, distributions)
2. Factor analysis to group like questions
3. Cluster analysis to segment customers
4. GE matrix to select target segment
5. Perceptual map to position
6. Discriminant analysis to find target customers
7. Classification analysis to apply discriminant function to new list of customers

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EMBA Program Example

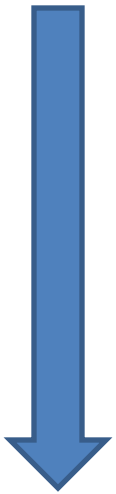


- You are hired to develop a marketing program to increase enrollment in the EMBA program
- What do you do (art or science)?
- Process steps (this is like your project for class)
  1. Qualitative interviews
  2. Quantitative survey
  3. STP analyses (cluster, discriminant, GE matrix, positioning map)
  4. Develop and implement plan (4Ps)

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Qualitative Questions: Probing for Important Attributes

general



specific

- Please discuss the process of deciding to get an Executive MBA.
- How did you begin your search for an Executive MBA program?
- What were important factors you considered when deciding on a specific EMBA program, and why were these factors important?
- What made you choose the Executive program over the full-time program?
- Of the schools you considered, what were the most compelling qualities of top choices?

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Segmentation Questions

- Key attributes when making a decision (needs and benefits)
- <u>All potential customers</u> report on importance of attributes

Please rate your agreement with the following statements:	**Strongly Disagree**						**Strongly Agree**	***construct***
A low-cost MBA program is very important to me.	1	2	3	4	5	6	7	*cost*
It is very important that an MBA program have a rigorous curriculum.	1	2	3	4	5	6	7	*rigor*
An easily accessible MBA program is important to me.	1	2	3	4	5	6	7	*location*
I think it is very important that an MBA program fits with my schedule.	1	2	3	4	5	6	7	*convenience*
It is very important that an MBA program is prestigious.	1	2	3	4	5	6	7	*prestige*
I feel it is very important that I am challenged by the coursework of an MBA program.	1	2	3	4	5	6	7	*rigor*
It is important that an MBA program is not expensive.	1	2	3	4	5	6	7	*cost*
I think it is very important that an MBA program be highly respected.	1	2	3	4	5	6	7	*prestige*

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Discriminant/Classification Questions



- Key demographics questions

What is your age?	_____
Please indicate your gender:	__male __female
What is your annual salary, including bonuses and commissions in US dollars?	__$0 - $25,000 __$25,001 - $50,000 __$50,001 - $75,000 __$75,001 - $100,000 __$100,001 - $125,000
How many miles is your home from the UW campus?	_____
How many years of work experience do you have?	
What was your undergraduate major?	_____

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Data Needed For Target Market Selection



- Segment Attractiveness
  - Growth rate of segments
  - Price sensitivity of segment
- Competitive Strength
  - UW relative advantage
  - Fit to UW brand image
- Where does this data come from?
  - Survey
  - Secondary sources
  - SWOT and 3Cs

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Perceptions and Preference Questions for Positioning Maps

- Across key attributes what do they think of you and the competitors and preference
- Can do on <u>all potential customers</u> or <u>just target customers</u>

Please evaluate each University's EMBA program (repeat for UW, Berkeley, etc):

Measure perceptions…

	Strongly disagree						**Strongly Agree**
Program is low cost	1	2	3	4	5	6	7
This is a high rigor program	1	2	3	4	5	6	7
The location is very good	1	2	3	4	5	6	7
Program is very convenient	1	2	3	4	5	6	7
Program is very prestigious	1	2	3	4	5	6	7

Measure preferences…

	low						**high**
Likelihood of selecting	1	2	3	4	5	6	7

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Segmentation: Create a Picture of Each Cluster

Segmentation variable / Cluster	Overall	Cluster 1	Cluster 2	Cluster 3
Low cost	4.8	6.67	1.21	6.79
High rigor	3.28	1.33	6.93	1.29
Good location	3.2	1.33	1.21	6.79
Very convenient	2.9	6.75	1.21	1.29
High prestige	5.2	1.33	6.93	6.79

Discriminant variable / Cluster	Overall	Cluster 1	Cluster 2	Cluster 3
Age	29.1	23.667	35.857	27
Gender	1.325	1.167	1.286	1.5
Salary	3.9	2.833	5.214	3.5
Miles from Campus	7.7	3.25	13.214	6
Work Experience	6.85	2.083	12.286	5.5

Cluster 1 ("young professionals") is concerned with cost and convenience and is younger with lower income and less job experience.

Cluster 2 ("returning executives") is primarily concerned with the rigor and prestige of the EMBA program. These customers tend to be older with higher salaries and more work experience.

Cluster 3 ("busy manager") wants a prestigious program with a close location at a reasonable price. They've got a little more experience and are making a reasonable salary.

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Targeting Using GE Matrix

Horizontal Axis



Competitive Strength	Young Professional	Returning Executive	Busy Manager
Fit to UW strengths	1.5	4.5	3
Fit to UW brand image	1.5	4	4

Vertical Axis



Segment Attractiveness	Young Professional	Returning Executive	Busy Manager
Growth rate of segment	2	4	3
High margins	2	4.5	1

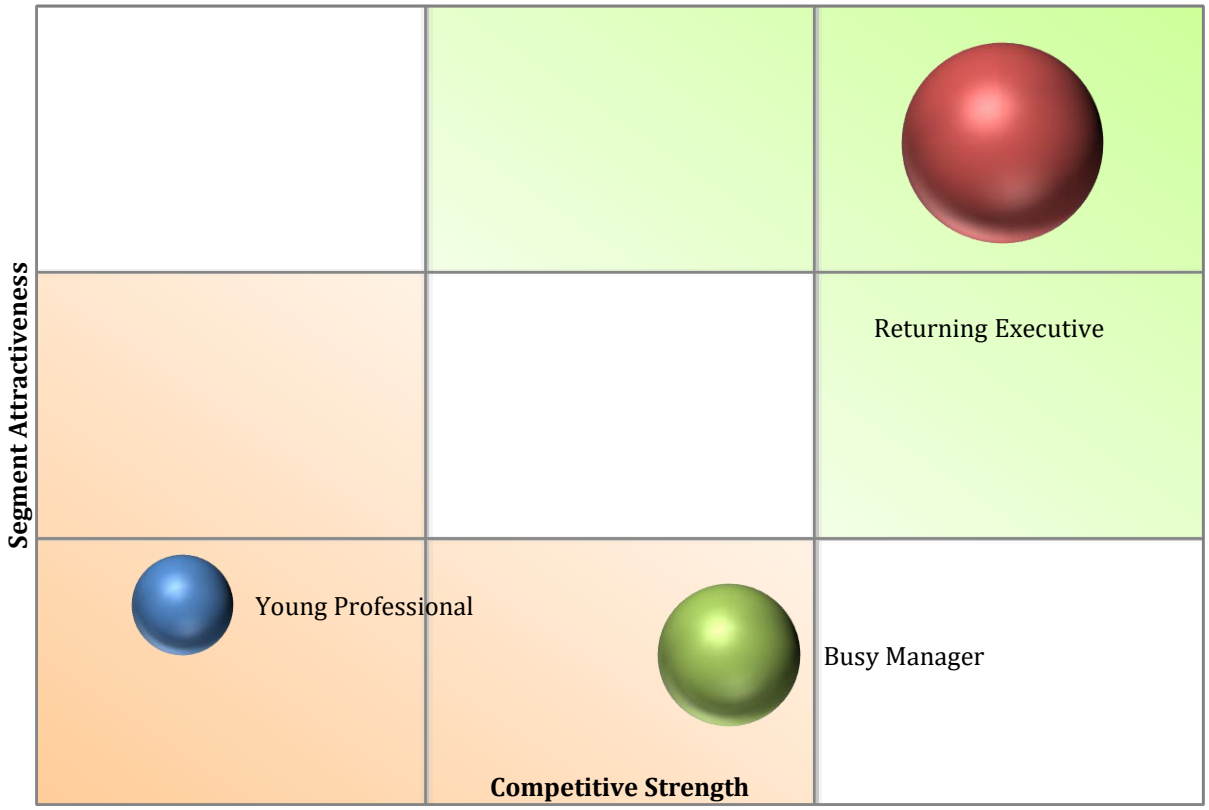
Bubble Size



	Young Professional	Returning Executive	Busy Manager
Market Size	5	20	10

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Which Segment Should You Target?



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Chapter 2 Book Case: Managing Customer Heterogeneity at DentMax



- DentMax is the market leader in digital imaging technology for dentists
- Continuously ranked first for image quality and diagnostic flexibility, but over the past decade, has lost 10% of market share
- **Problem:** understand how the needs, desires, and preferences across dentists differ in the entire industry
- **Data:** RFID tracking data from DENTEXPO trade show (time spent at each booth)→ combined with past purchase behavior, X-ray imaging product needs and media trust (from post-expo survey)
  - Cluster analysis identified 4 clusters:
    - Loyalists (12%)
    - Switchables (22%)
    - Generalists (49%)
    - Apathetics (17%)

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Analytics Driven Case: Managing Customer Heterogeneity at DentMax



- Targeting and Positioning for Competitive Advantage
  - Loyalists – very appealing
    - Value image quality and diagnostic flexibility, DentMax's strengths → further correspond with dentists through magazine journals, and use peer feedback
  - Switchables – good option to grow market share
    - Value software integration and technical assistance highly → stress these qualities to dentists
    - Care more about information from internet → use Internet marketing
  - Generalists – not a good option
    - Valued price and technical assistance highly → DentMax doesn't want to compete on these dimensions
  - Apathetics – not appealing

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